Aluminium-Enhanced Detonation Simulation

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Abstract

This report describes the development of a simulation capability for aluminised high explosives, from detonation of the charge to afterburning in air, built on the open-source `blastFoam` solver suite (OpenFOAM-12). The target problem is a Composition B charge with 30% aluminium by mass (Comp-B/Al). The modelling uses the multi-fluid solver `blastEulerFoam`, with separate phases for the detonation products, the aluminium particles and the surrounding reactive gas. The report sets out the physical problem and the models chosen for each part of it, distinguishes the capability that existed in `blastFoam` from what was added or changed in this work, describes the verification performed so far, and assesses the status against the aims. At present the detonation and early product expansion are modelled and verified in one dimension; the transfer of products to a reactive gas phase is verified in isolation but not yet used in the Comp-B/Al cases, and the aluminium heating, ignition and combustion models exist only as standalone reference models. No comparison with experimental data for aluminised charges has been made yet.

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1 Introduction

1.1 Aim

The aim of this work is a validated simulation method for the detonation of an aluminised explosive charge and the subsequent blast and afterburning in air. The process has four parts, each with its own physics and time scale:

1. detonation of the charge and early expansion of the detonation products, governed by a high-pressure equation of state (microseconds);
2. mixing of the products with the surrounding air and finite-rate combustion of the fuel-rich products (CO, H_2, ...) in this mixture (tens of microseconds to milliseconds);
3. transport, heating, ignition and combustion of the aluminium particles dispersed in the expanding gas (hundreds of microseconds to milliseconds);
4. afterburning of the aluminium in the mixture of products and air, which raises the late-time pressure and the positive impulse.

The final target is a multi-dimensional simulation of a Comp-B/Al charge whose peak overpressure, arrival time and positive impulse agree with free-field gauge data (within 8%, Section 10).

1.2 Approach

The work builds on `blastFoam` [1], an OpenFOAM-based suite for high-explosive detonation and airblast. Two of its solvers are relevant. `blastFoam` solves multiphase flows with a single mixture velocity and energy; `blastEulerFoam` is a multi-fluid (Euler–Euler) solver in which each phase has its own mass, momentum and energy equations, coupled through finite-rate exchange terms. Since the aluminium particles lag the gas, heat up at their own rate and burn after a delay, they are modelled as a separate phase, and the work therefore uses `blastEulerFoam`. The development is staged from zero-dimensional (0D) unit tests through one-dimensional (1D) planar and spherical cases to multi-dimensional simulations, with an automated test suite at each level (Section 8).

External tools are used for reference data: the ideal detonation code `EuDetCode` (chemical equilibrium, CJ state, expansion isentropes and JWL fits; a separate code, not part of the `blastFoam` repository) and the 1D multiphase code `gunIB1D` for cross-solver comparison of granular shock problems.

1.3 Structure of the report

Section 2 describes the multi-fluid modelling framework and lists what was pre-existing and what was developed in this work. Sections 3–6 cover the physical models for the four parts of the problem: the explosive and its products, the transfer of the products to a reactive gas, the aluminium particles, and mixing and afterburning. Section 7 describes the numerical methods, including the changes needed to make the multi-fluid solver conservative and stable for this problem. Section 8 describes the verification approach and Section 9 the results. Section 10 assesses the status against the aims and gives the plan for the remaining work. Throughout, [UPSTREAM] marks capability that existed in `blastFoam` before this work, and [THIS WORK] marks what was added or changed.

2 Modelling Framework

2.1 Phases

A Comp-B/Al charge is modelled with three phases:

- `compBAl`: the explosive and its detonation products, a 'detonating' fluid phase in which a reaction progress variable λ moves the state from the unreacted explosive (Murnaghan EOS) to the products (`eJWLC` EOS, Section 3);
- `Al`: the aluminium particles, a granular phase with a compressible solid EOS (Murnaghan) and kinetic theory for the particle stresses;
- `gas`: the surrounding air and, once transferred, the detonation products as a multicomponent reactive gas (NASA-9 thermodynamics in the reacting cases, JANAF in the inert ones; Abel–Nobel EOS; Section 4).

Initially the charge region holds $\alpha_{\mathrm{compBAl}} = 0.818$ and $\alpha_{\mathrm{Al}} = 0.182$ (27% of the charge mass as inert Al at $1820\,\mathrm{kg/m^3}$; Section 3.1), and the gas fills the rest of the domain.

2.2 Governing equations

`blastEulerFoam` [UPSTREAM] solves, for each phase k with volume fraction α_k, density ρ_k, velocity $\mathbf{u}_k$, pressure p_k and specific total energy $E_k = e_k + \frac{1}{2}|\mathbf{u}_k|^2$, a set of balance equations of Baer–Nunziato type [3, 4]:

$$\frac{\partial \alpha_k \rho_k}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k) = \Gamma_k, \tag{1}$$

$$\frac{\partial \alpha_k \rho_k \mathbf{u}_k}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{u}_k \mathbf{u}_k + \alpha_k p_k \mathbf{I}) = p_I \nabla \alpha_k + \mathbf{M}_k + \Gamma_k \mathbf{u}_I, \tag{2}$$

$$\frac{\partial \alpha_k \rho_k E_k}{\partial t} + \nabla \cdot \big(\alpha_k (\rho_k E_k + p_k) \mathbf{u}_k\big) = -p_I \frac{\partial \alpha_k}{\partial t} + \mathbf{M}_k \cdot \mathbf{u}_I + Q_k + \Gamma_k E_I, \tag{3}$$

where Γ_k is the interphase mass transfer, $\mathbf{M}_k$ the interphase momentum transfer (drag), Q_k the interphase heat transfer, and p_I, $\mathbf{u}_I$, E_I are interfacial values set per phase pair (`interfacialPressure`, `interfacialVelocity`). The volume fractions sum to one; the fluid volume fractions are advected with the mixture velocity, and the granular phase adds the kinetic-theory stresses (granular temperature Θ, Lun et al. collisional pressure, Johnson–Jackson frictional stress [11]). Species mass fractions in the gas are transported with the gas mass flux and react through finite-rate chemistry (Section 4.3).

2.3 Interphase coupling

The phases exchange momentum and energy through finite-rate models and through relaxation steps applied after each stage of the time integration:

- *Particle drag and heat transfer* [UPSTREAM]: Gidaspow–Ergun–Wen–Yu drag and Ranz–Marshall heat transfer between the Al and each fluid phase. None of the available drag laws depends on the particle Mach number (Section 5.3).
- *Products/gas velocity*: the two fluid phases are relaxed instantaneously to a common velocity (`VRelaxation instant`) [UPSTREAM]; the energy update was made conservative [THIS WORK] (Section 7.2).
- *Products/gas pressure and temperature* [THIS WORK]: in cells where the products and the gas are numerically mixed, the volumes of the non-carrier fluid phases are exchanged until the pressures agree (`equilibrateDiluteVolumes`, with the $\bar{p}\,\Delta\alpha$ work moved between the phase energies), and the temperatures are relaxed by a stiff constant-coefficient heat transfer integrated exactly. The upstream pressure-relaxation solver collapses the time step in these cells and is not used.
- *Mass transfer*: products → gas by the `pressurePhaseChange` model [THIS WORK] (Section 4); Al → gas by the reacting-particle model with an empirical burn rate [UPSTREAM], limited to gas-phase oxidiser [THIS WORK] (Section 5.3).

2.4 Pre-existing capability and contributions of this work

The work is on a fork of `blastFoam`; all changes are relative to the upstream commit `71b97fd7`, and are listed in detail in `UPDATES.md` of the repository. Table 1 summarises the division.

Table 1: Capability used in this work: pre-existing in `blastFoam` (upstream `71b97fd7`) and added or changed in this work.

Area	**Pre-existing in `blastFoam`**	**Added or changed in this work**
Solvers	`blastFoam`, `blastEulerFoam` (multi-fluid, kinetic theory, drag and heat-transfer models, relaxation), `blastParcelFoam` (Lagrangian), `blastReactingFoam`	Conservative phase decode; dilute-phase volume equilibration; energy-conserving velocity relaxation and drag work; granular phase transports thermal energy (with compression energy) instead of total energy; mixture-pressure buoyancy on the granular phase; no density floor in the fluid decode
Detonation	JWL, JWLC, BKW and other EOS; `programmedIgnition` activation; detonating phase thermo	`eJWLC` EOS (3-term isentrope, energy-based, T_{CJ}/ω_T temperature closure); `nodeProgrammed` activation; products-only fitting procedure and E_0/ω_T calibration
Reactive gas	JANAF thermo, Abel–Nobel EOS, multicomponent mixtures, chemistry and combustion models (single-phase systems)	Multicomponent and reacting thermo for individual phases of a multiphase system; JANAF extrapolation with constant c_p above the fit range; mass-weighted covolume mixing
Phase transfer	Temperature-based transfer was not available in this code base	`pressurePhaseChange` (pressure-interval transfer with species allocation; conservative mass, momentum and energy transfer in `blastEulerFoam`)
Aluminium	Reacting diameter model with empirical rates (pressure-based, Arrhenius); reacting-particle mass transfer	Oxidiser limit on the burn rate; registered initial diameter (fixes AMR crash); Al gas-phase reactions in the test cases; standalone reference models for heating, ignition, d^n combustion and condensation (not in the solver)
Numerics	HLL, HLLC, AUSM+, Kurganov–Tadmor flux schemes; MUSCL reconstruction; RK time integration; AMR and load balancing	HLLC phase flux rewritten to the standard star-state form; trace-phase mixture impedance (`alphaTrace`); covolume and temperature-range guards; Courant number with mass exchange
Post-processing	Probes, field function objects	`temporalMax`, `temporalIntegral` (peak pressure and impulse maps, AMR-aware)
Testing	Tutorials (no automated verification)	0D/1D/1D-spherical test cases with verification scripts; regression runner (`bin/run_blast_tests.py`); EOS fitter (`utilities/thermo_fitter`); report figure generation

3 Explosive Charge and Detonation Products

3.1 Composition and treatment of the aluminium

In an aluminised explosive the charge is a mixture of the high explosive and aluminium particles. Micron-sized particles ($d \gtrsim 1\,\mu$m) react little in the detonation zone ($\sim$ 10–100 ns); they act as a momentum and heat sink that lowers the detonation velocity and pressure, and burn later in the expanding products and in air [13]. Nano-sized particles can react partly in the early expansion.

The inert part of the aluminium is modelled as a separate granular phase. This lets the particles develop their own velocity and temperature, which matters for the ignition delay and the spatial distribution of the later energy release. It also means that the products EOS must describe the products alone: the thermochemical calculation includes the aluminium in the reactants, but the JWL-type fit is made to the products isentrope with the volume of the inert aluminium excluded, so that the aluminium is not counted twice. For the Comp-B/Al charge (70% Comp-B, 30% Al), 90% of the aluminium is taken as inert in the detonation, i.e. 27% of the charge mass, giving $\alpha_{\mathrm{Al}} = 0.182$ and a products-phase initial density $\rho_0 = 1624.45\,\mathrm{kg/m^3}$.

3.2 Programmed burn

The detonation is not resolved; the reaction front is prescribed (programmed burn). In the `nodeProgrammed` activation model [THIS WORK], the arrival times $t = t_0 + |\mathbf{x} - \mathbf{x}_{\mathrm{ign}}|/D_{\mathrm{CJ}}$ at the vertices of each cell give a start and end time t_{start}, t_{end}, and the progress variable ramps linearly across the cell,

$$\lambda = \min\left(1, \max\left(0, \frac{t - t_{\mathrm{start}}}{t_{\mathrm{end}} - t_{\mathrm{start}}}\right)\right), \tag{4}$$

optionally also triggered by compression ahead of the front (`PROGRAMMEDBETA`). The detonating phase blends the unreacted and products states with λ. Because the front speed is imposed, a correct detonation velocity in a simulation is not a test of the model; what can be tested is the pressure and flow behind the front, which depend on the EOS and on the resolution (Section 9.5). The detonation velocity D_{CJ} and pressure p_{CJ} come from `EuDetCode`.

3.3 The eJWLC equation of state

The products are described by the energy-based Mie–Grüneisen form `eJWLC` [THIS WORK], built on the three-term (JWLC) isentrope through the CJ state. With $V = \rho_0/\rho$,

$$p_s(V) = A\,e^{-R_1 V} + B\,e^{-R_2 V} + C\,V^{-(\omega+1)}, \tag{5}$$

$$e_s(V) = v_0\left[\frac{A}{R_1}e^{-R_1 V} + \frac{B}{R_2}e^{-R_2 V} + \frac{C}{\omega}V^{-\omega}\right] + E_0, \tag{6}$$

$$p(\rho, e) = p_s(V) + \omega\rho\,\big(e - e_s(V)\big), \tag{7}$$

where e_s follows from $\mathrm{d}e_s = -p_s\,\mathrm{d}v$ along the isentrope, $v_0 = 1/\rho_0$, and ω is the (constant) Grüneisen coefficient. The constant E_0 sets the energy datum; it does not change $p(\rho, e)$ for given $e - e_s$, and is used to put the products on the same datum as the gas phase (Section 4.4).

JWL-type equations are mechanical: they relate p, v and e but do not define a temperature. A temperature is needed for the heat transfer to the particles and to the gas. `eJWLC` uses a power law along the isentrope and a constant heat capacity off it:

$$T_s(V) = T_{\mathrm{CJ}}\left(\frac{V_{\mathrm{CJ}}}{V}\right)^{\omega_T}, \qquad T(\rho, e) = T_s(V) + \frac{e - e_s(V)}{C_v}, \tag{8}$$

with T_{CJ} and $V_{\mathrm{CJ}} = \rho_0/\rho_{\mathrm{CJ}}$ from `EuDetCode`. The exponent ω_T defaults to ω; since it does not enter the pressure, it can be chosen so that T_s also passes through a second known state, here the temperature of the gas formed at the transfer pressure (Section 4.4). With $\omega_T \neq \omega$ the closure is not thermodynamically consistent with a constant-C_v Mie–Grüneisen EOS; it is a temperature closure only.

3.4 Parameters for Comp-B/Al

The parameters are fitted to the CJ state and expansion isentrope from `EuDetCode` (BKW EOS, `Gas24CHNO` database, reactants RDX 0.42, TNT 0.28, Al 0.30 by mass, 90% of the Al inert). Table 2 lists the fit to the products alone, used in all Comp-B/Al cases, and the fit to the full mixture for comparison; Figure 1 shows the isentropes. The products fit is within 5% of the equilibrium isentrope below 2 GPa (3.8% at the 100 bar transfer pressure, −5.8% at 20 bar) but up to 15% high near 9 GPa; the full-mixture fit has errors of similar size. The products fit gives $D_{\mathrm{CJ}} = 7154.23\,\mathrm{m/s}$, $p_{\mathrm{CJ}} = 20.15\,\mathrm{GPa}$, $T_{\mathrm{CJ}} = 3844.39\,\mathrm{K}$ and $\rho_{\mathrm{CJ}} = 2150.55\,\mathrm{kg/m^3}$. In the cases, $C_v = 1450\,\mathrm{J/(kg\,K)}$ (JANAF c_v of the frozen product gas at 2500 K), $E_0 = -503.47\,\mathrm{kJ/kg}$ and $\omega_T = 0.3711$ (Section 4.4).

Table 2: eJWLC parameters for 70% Comp-B / 30% Al (90% of the Al inert) fitted to the `EuDetCode` isentrope. The products fit is used in the simulations, with the C_v, E_0 and ω_T of Section 3.4.

Parameter	Full mixture	Products only	Unit
ρ_0	1820	1624.45	$\mathrm{kg/m^3}$
A	1428.3	224.94	GPa
B	12.17	2.773	GPa
C	1.133	1.470	GPa
R_1	5.858	3.456	–
R_2	1.386	0.8134	–
ω	0.3013	0.3027	–

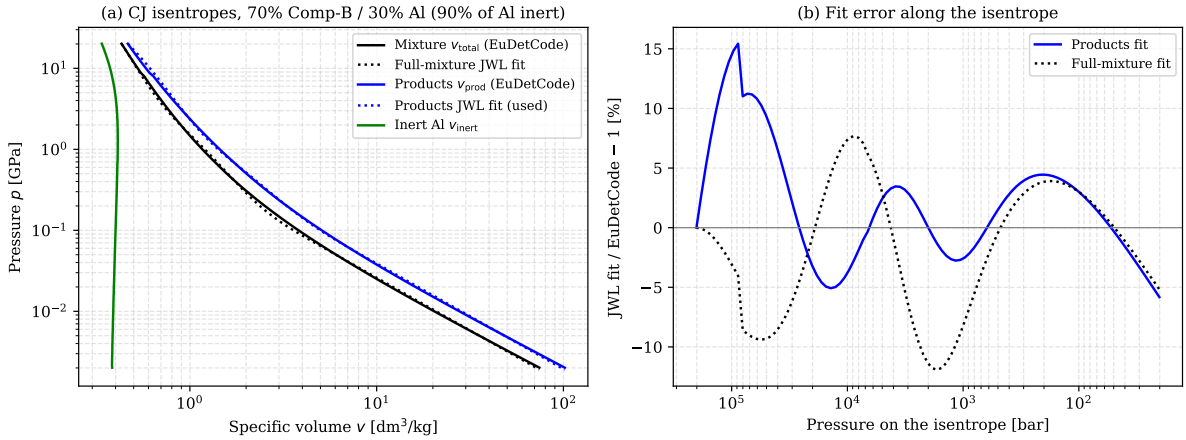


Figure 1: CJ expansion isentropes from `EuDetCode` for 70% Comp-B / 30% Al (90% of the Al inert), down to 20 bar: (a) full mixture (v_{total}), products (v_{prod}) and inert aluminium (v_{inert}), with the full-mixture and products JWL fits; (b) relative error of the fits along the isentrope.

3.5 Limitations

- *Range.* The fit covers expansion to $V/V_{\mathrm{CJ}} \approx 10$–$20$. Beyond that the constant-ω tail is not reliable, which is one reason for transferring the products to a gas EOS (Section 4).
- *Temperature.* The power law with constant exponent does not follow the equilibrium isentrope temperature: along the expansion the equilibrium temperature falls faster, and the power law with $\omega_T = \omega$ is 550–820 K too high between 0.1 and 1 GPa (Figure 2). With the calibrated ω_T the error is reduced to 270–630 K there and changes sign near the transfer pressure (−80 K at 100 bar, −190 K at 20 bar). The constant C_v ignores shifting equilibria and vibrational excitation. Temperature errors of this size affect the heat transfer to the particles and therefore the ignition delay.

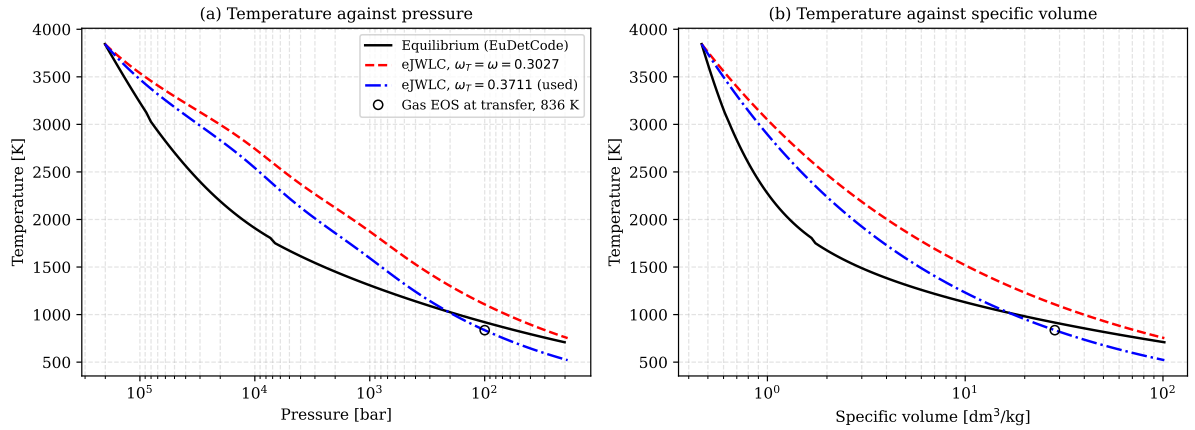


Figure 2: Temperature along the CJ isentrope of the products of 70% Comp-B / 30% Al: equilibrium (`EuDetCode`) and the eJWLC power law with $\omega_T = \omega$ and with the calibrated ω_T used in the cases, against (a) pressure and (b) specific volume. The circle is the gas-EOS temperature at the 100 bar transfer state.

4 Transfer of the Products to a Reactive Gas Phase

The detonating phase does not carry species. To model combustion of the fuel-rich products with air, the products are transferred, once expanded, to the multicomponent gas phase with a frozen product composition.

4.1 Transfer criterion

The transfer is triggered by pressure rather than temperature. The eJWLC pressure follows the equilibrium isentrope to within about 5% between 2 GPa and 20 bar for this fit, while its temperature is in error by hundreds of kelvin (Section 3.5), so a temperature threshold would trigger at the wrong state. A pressure window also corresponds to the physical freeze-out of the product chemistry, which occurs at roughly 50–200 bar as the expansion becomes faster than the recombination reactions.

4.2 The `pressurePhaseChange` model

The `pressurePhaseChange` model [THIS WORK] transfers mass from a source phase (the products) to one or more target phases at the rate

$$\dot{m} = \frac{\alpha_s \rho_s}{\tau}\,\Gamma(p)\,\lambda_s, \tag{9}$$

where τ is a relaxation time, λ_s the burn fraction of the source, and $\Gamma(p)$ a smooth window between p_{lower} and p_{upper} of width p_{range},

$$\Gamma(p) = \tfrac{1}{4}\left(1 + \tanh\frac{p_{\text{upper}} - p}{p_{\text{range}}}\right)\left(1 + \tanh\frac{p - p_{\text{lower}}}{p_{\text{range}}}\right), \tag{10}$$

cut off two widths outside the window (a one-sided form is used for $p_{\text{lower}} = 0$). The transferred mass is distributed over the target species with prescribed mass or mole fractions (the frozen `EuDetCode` composition at freeze-out). In `blastEulerFoam` the transferred mass carries the source velocity and specific total energy, and the volume moves with the mass; mass, momentum and total energy are then conserved by the transfer itself. The energy datums of the two phases must agree; this is arranged offline through E_0 (Section 4.4), and no state-dependent energy correction is applied at run time.

4.3 The gas phase

The gas phase is a multicomponent mixture of air and product species. In the reacting Comp-B/Al cases it has 24 species (O_2, N_2, CO, CO_2, H_2O, H_2, NH_3, CH_4, C_2H_6, C(gr), Al, Al_2O_3(s) and the dissociation and intermediate species O, N, NO, H, OH, AlO, HO_2, H_2O_2, CH_3, CH_2O, HCO, CH_3O) with NASA-9

thermodynamics to 20 000 K; the inert cases use the first twelve with JANAF 7-coefficient thermodynamics. Transport is Sutherland and the EOS Abel–Nobel (covolume)

$$p = \frac{\rho R_{\mathrm{mix}} T}{1 - b\rho}, \qquad b = \sum_k Y_k b_k, \tag{11}$$

which accounts for the excluded volume of the dense product gas and tends to the ideal gas for $b\rho \ll 1$. Species are transported with the gas mass flux. Above $T_{\mathrm{eq}} = 2500\,\mathrm{K}$ the gas is set to local chemical equilibrium at its density and internal energy (element-potential solver, condensed species as pure phases); below, it reacts by finite-rate chemistry (a reversible GRI-3.0 $H_2/O_2/CO/C1$ subset, 68 reactions, and global Al oxidation steps; laminar, integrated with a stiff ODE solver). The multicomponent and reacting thermo for an individual phase of a multiphase system was added in this work; the underlying JANAF, Abel–Nobel and chemistry models are upstream.

High temperatures. The JANAF fits of the air species end at 5000 K. In `blastFoam` the properties were evaluated at the clamped temperature, so the gas could not exceed 5000 K and the energy of strongly shocked air was misrepresented. In this work the properties are extrapolated with constant c_p above the fit range, with enthalpy and entropy extended consistently, and the decode temperature cap is raised to 50 000 K. This removes the cap but not the physics that is missing: there is no dissociation, so strongly shocked air is too hot (up to 14 400 K in the spherical cases and 24 800 K in a planar case, where equilibrium air would stay near 10 000 K or below). NASA-9 thermodynamics to 20 000 K (`nasa9` thermo, species N_2, O_2, NO, N, O, H, OH and the product species) and local chemical equilibrium above $T_{\mathrm{eq}} = 2500\,\mathrm{K}$ in the gas chemistry (element-potential solver at fixed (e, ρ), verified against `EuDetCode`) are now available; the shock-in-air test uses them (Section 9.2), and the reacting Comp-B/Al cases use them too (Section 9.5).

4.4 Reconciliation of the two equations of state

The two phases measure energy from different datums. The JANAF datum includes the heats of formation, so the frozen product gas has a large negative energy; the eJWLC datum is arbitrary (the constant E_0). If the datums are not reconciled, mass transferred at constant specific energy arrives in the gas at the wrong temperature and pressure; with $E_0 = 0$ the pressure after transfer at 100 bar would be about 50% too high.

The matching is done at the transfer pressure $p_{\mathrm{tr}} = 100\,\mathrm{bar}$, on the products isentrope ($v_{\mathrm{tr}} = 2.838 \times 10^{-2}\,\mathrm{m^3/kg}$):

1. *Pressure continuity.* The gas temperature at $(p_{\mathrm{tr}}, v_{\mathrm{tr}})$ follows from the Abel–Nobel EOS of the frozen composition ($R_{\mathrm{mix}} = 324.35\,\mathrm{J/(kg\,K)}$, $b = 1.263 \times 10^{-3}\,\mathrm{m^3/kg}$): $T_{\mathrm{tr}} = p_{\mathrm{tr}}(v_{\mathrm{tr}} - b)/R_{\mathrm{mix}} = 836.10\,\mathrm{K}$.
2. *Energy datum.* E_0 is chosen so that the eJWLC energy at v_{tr} equals the gas energy at T_{tr} on the `blastFoam` gas convention, $e_{\mathrm{gas}} = \sum_k Y_k(h_{s,k} - R_k T) = 0.4341\,\mathrm{MJ/kg}$; this gives $E_0 = -503.47\,\mathrm{kJ/kg}$.
3. *Temperature continuity.* ω_T is chosen so that $T_s(V_{\mathrm{tr}}) = T_{\mathrm{tr}}$ with T_{CJ} unchanged:

$$\omega_T = \frac{\ln(T_{\mathrm{CJ}}/T_{\mathrm{tr}})}{\ln(V_{\mathrm{tr}}/V_{\mathrm{CJ}})} = 0.3711.$$

The procedure is implemented in `utilities/thermo_fitter/products_only_fit.py` [THIS WORK]. It gives exact pressure and temperature continuity at the matching state; elsewhere in the transfer window the mismatch depends on how closely the two EOS agree. A further step, matching the slope of the isentrope (C^1 continuity) by setting the asymptotic ω to $\gamma_{\mathrm{mix}} - 1$ of the frozen gas, is implemented in the fitter (`fit_ejwlc_nobleabel.py`) but is not used in the current fit, which keeps the `EuDetCode` value of ω. Figure 3 compares the two descriptions along the expansion.

A state-dependent energy correction (e.g. the difference between the two EOS energies at the local temperature) is deliberately not used: it adds energy in proportion to the JWL temperature error. In a 0D test it added about 480 kJ/kg and raised the pressure by 56%.

4.5 Status

The transfer is verified in a 0D reactor and in a planar detonation (Section 9). Since 2026-10-14 it is enabled in the reacting and Al-combustion spherical cases (below 100 bar, $\tau = 20\,\mu\mathrm{s}$, with a stiff `seulex` chemistry integrator; the explicit RKCK45 integrator failed on the Al reactions); the products then burn with the air (Section 9.5). It is not yet enabled in the tutorials.

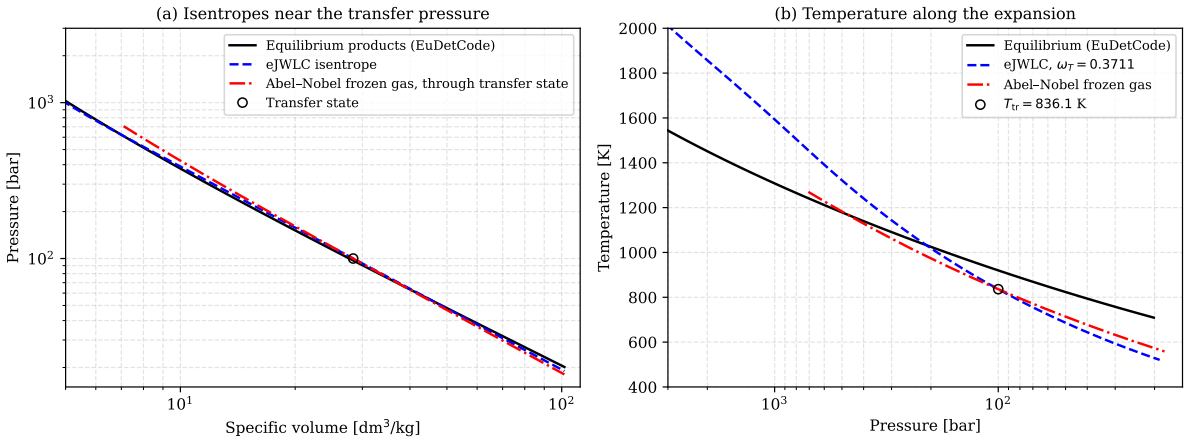


Figure 3: Products and gas around the transfer state for 70% Comp-B / 30% Al: (a) isentropes of the equilibrium products (`EuDetCode`), the eJWLC fit, and the Abel–Nobel frozen gas through the 100 bar transfer state; (b) temperatures along the expansion (eJWLC with the calibrated ω_T). The pressure error of the fit along the isentrope is shown in Figure 1(b).

5 Aluminium Particles

5.1 Physical processes

Heating, melting and ignition. The particles are heated by convection from the products and the shocked air. Aluminium melts at 933 K ($\Delta H_{\text{fus}} \approx 397$ kJ/kg), which holds the particle temperature during melting. The particles are coated with an oxide shell that prevents combustion until it fails, at about 2050–2300 K near the oxide melting point; the ignition temperature depends on the heating rate and the environment. At the heating rates behind a blast wave ($h \sim 10^4$–10^5 W/(m^{2}K)) the particle interior lags its surface, which ignites first.

Combustion. Micron-sized particles burn with a detached vapour-phase diffusion flame; the burn time scales as $t_b \propto d^n$ with $n \approx 1.8$–2. Nano-sized particles burn by surface kinetics with $n \approx 1$. A widely used correlation for micron particles in mixed oxidisers is that of Beckstead [13],

$$t_b\,[\text{ms}] = \frac{a\,d^{\,n}}{X_{\text{eff}}\,p^{0.1}\,T_0^{0.2}}, \qquad X_{\text{eff}} = X_{\text{O}_2} + 0.6X_{\text{H}_2\text{O}} + 0.22X_{\text{CO}_2}, \tag{12}$$

with $a = 0.00735$, $n = 1.8$, d in μm, p in atm and the ambient temperature T_0 in K. For the 20–40 μm powders of typical aluminised explosives the correlation gives 1.6–5.5 ms in air at 2800 K, so the aluminium is essentially unburnt during the detonation and early expansion and releases its energy in the fireball.

Reactions and energy. The aluminium reacts with O_2 (31 MJ/kg Al to Al_2O_3), H_2O (15.4 MJ/kg) and CO_2 (13.6 MJ/kg). The reactions with H_2O and CO_2 allow burning inside the products before air is entrained. The flame forms gaseous sub-oxides and Al_2O_3 vapour, which condense to sub-micron oxide smoke; the condensation (about 18–19 MJ/kg of oxide) is a large part of the heat release and holds the flame temperature near the oxide boiling point (about 3800–4000 K at elevated pressure). The smoke is a strong emitter, so radiation can be a significant loss from the fireball (Section 6.3).

5.2 Modelling approaches

Four approaches are candidates for the particle phase; their selection is part of the planned work (Section 10).

- *Eulerian multi-fluid* (current): the particles are a continuum phase with its own velocity and temperature. Appropriate in the dense charge ($\alpha_{\text{Al}} \approx 0.2$) and early expansion; a single size class per phase; numerical diffusion at the edge of the dispersing cloud.

- *Lagrangian parcels* (`blastParcelFoam` [UPSTREAM]): particle trajectories with their own size, temperature and burn history; natural polydispersity and no numerical diffusion. Standard implementations neglect the particle volume, which is not acceptable in the dense charge; a dense formulation (gas volume fraction in the carrier equations, particle stress, kernel projection of parcel volume) would be needed.

- *Population balance*: size classes (sectional) or transported moments (QMOM) of the size distribution in the Eulerian framework, with burning as a moment source.

- *Hybrid*: Eulerian in the dense core, converted conservatively to Lagrangian parcels where α_{Al} falls below a threshold (about 10^{-3}).

5.3 Current implementation

In the solver the aluminium is a granular Eulerian phase [UPSTREAM] with a compressible solid EOS (Murnaghan), Gidaspow–Ergun–Wen–Yu drag and Ranz–Marshall heat transfer. The burn rate of the reacting diameter model [UPSTREAM] is the empirical pressure law $r_b = a\,(p/1\,\mathrm{MPa})^n$ for $p \geq p_{\min}$ ($a = 5 \times 10^{-5}$ m/s, $n = 0.7$, $p_{\min} = 2$ bar, $d_0 = 30\,\mu$m), chosen to give a burn time of the order of 0.5 ms. This law is not a physical model of aluminium combustion: it has no ignition delay and no oxidiser dependence. This work added a limit to gas-phase oxidiser (the rate is multiplied by the gas fraction of the surrounding fluid and by $\min(1, \sum Y_{\mathrm{ox}}/Y_{\mathrm{ref}})$ over O_2, CO_2, H_2O), so that the particles do not burn inside the products, and a correction so that the granular phase transports its thermal energy with the compression energy of the solid EOS (before, the particles cooled to the 100 K floor behind the detonation).

Added to the solver in Phase D, each with a 0D test (Section 9.6), but not yet used in the Comp-B/Al cases:

- *Melting* (`eMelting` thermodynamics): constant heat capacities of the solid (1059 J/(kg K), the mean between 298 and 933 K) and the liquid (1177 J/(kg K)) and the latent heat (396.6 kJ/kg) released over 2 K around 933.47 K (NASA/CODATA data). The energy is piecewise linear in temperature and is inverted exactly; Newton's method on the kinked curve can cycle between the branches.

- *Compressible and rarefied drag* (`Loth`): the correlation of Loth [14], a function of the Reynolds number, the particle Mach number $|U_r|/c$ and the Knudsen number $\sqrt{\pi\gamma/2}\,Ma/Re$, with a rarefaction branch (Re < 45) and a compression branch; at $Ma \to 0$ it reduces to Schiller–Naumann with the Clift–Gauvin term.

- *Rarefied heat transfer* (`Kavanau` Nusselt number): $Nu = Nu_c/(1 + 3.42\,Ma\,Nu_c/(Re\,Pr))$ with $Nu_c = 2 + 0.459\,Re^{0.55}Pr^{0.33}$ [15]. A 1 μm particle in air at 0.2 bar and 2500 K has $Nu = 0.28$ instead of 2. The Prandtl number of a phase pair was computed with c_v instead of c_p (fixed).

- *Ignition and Beckstead burning* (`Beckstead` diameter rate): particles ignite at T_{ign} and burn by the d^n law of Eq. (12), with X_{eff} from the gas composition; once burning, a particle continues when it is cooled below T_{ign}. Testing this found two errors in the reacting diameter model: the particle number density used the initial diameter, so a burnt-out cloud kept $1/e$ of its mass, and a volume floor of 10^{-15} m^3 limited the burning of particles below 12 μm (fixed).

Not in the solver: oxide condensation and the internal temperature profile of the particle (the reference model's surface ignition).

6 Mixing, Turbulence and Afterburning

6.1 Mixing of products, particles and air

The products and the aluminium can only burn with air after mixing with it. At the product–air contact the mixing is driven by the Richtmyer–Meshkov instability (shock passage across the density interface, including the reflected shocks), the Rayleigh–Taylor instability (deceleration of the dense products by the air) and Kelvin–Helmholtz instability of the resulting shear layers. At fireball temperatures the gas-phase reactions and the burning of small particles are fast compared with the mixing (Damköhler number $\gg 1$), so the afterburning rate is controlled by the mixing.

The gas and the particles mix differently. Gas-phase products follow the flow and mix down to molecular scales; their combustion is confined to the turbulent mixing layer. Particles have inertia, characterised by

the Stokes number $\mathrm{St} = \tau_p/\tau_f$ with $\tau_p = \rho_p d^2/(18\mu_g)$: large particles ($\mathrm{St} > 1$, $d \gtrsim 15$–$30\,\mu$m) decouple from the decelerating gas and penetrate ahead of the products into fresh air, small particles follow the gas, and particles with $\mathrm{St} \sim 1$ concentrate between vortices. The aluminium can therefore burn in a distributed volume ahead of the gaseous fireball, which sustains the late-time pressure.

These effects are multi-dimensional; in 1D the particle cloud stays inside the products except at its dilute edge (Section 9.5).

6.2 Turbulence and turbulence–chemistry interaction

Large-eddy simulation (LES, with e.g. WALE or dynamic Smagorinsky subgrid models) is the natural choice for instability-driven mixing, which is dominated by large coherent structures; RANS (k–ω SST with compressibility corrections) is cheaper for parametric studies in 2D. The mean reaction rate cannot be evaluated at the mean temperature and composition; subgrid closures such as the partially stirred reactor (PaSR, rate reduced by $\kappa = \tau_c/(\tau_c + \tau_{\mathrm{mix}})$) or the eddy dissipation concept account for the unresolved mixing. Turbulence models are available in OpenFOAM and `blastFoam` [UPSTREAM]; their use for this problem has not been assessed.

6.3 Thermal radiation

Radiation from the oxide smoke at 3000–4000 K can be a significant energy loss from the fireball. The radiative transfer equation can be solved in the P1 or discrete-ordinates approximation with a Planck-mean absorption coefficient of the smoke. Radiation models are available in `src/radiationModels` [UPSTREAM], but no solver used here couples them to the multi-fluid energy equations; this would be needed if the radiative loss turns out to be significant.

7 Numerical Methods

7.1 Discretisation

The equations are solved by a finite-volume Godunov method [UPSTREAM]. Each phase has its own approximate Riemann solver; the Comp-B/Al cases use HLLC [5] for all phases, with second-order MUSCL reconstruction (Minmod limiter) and a two-stage strong-stability-preserving Runge–Kutta scheme (RK2SSP). AUSM+, HLL and Kurganov–Tadmor are also available. The time step is limited by the phase Courant numbers (0.2 in the Comp-B/Al cases) and, in this work, by the rate of interphase mass loss. Adaptive mesh refinement (AMR) and load balancing are available [UPSTREAM] but not yet used in the Comp-B/Al cases.

This work rewrote the HLLC phase flux to the standard star-state form $F + S(U^* - U)$ with separate owner and neighbour star states, an explicit branch for left-moving waves and a guarded contact speed.

7.2 Multiphase numerics: conservation and trace phases

A multi-fluid solver must handle cells where a phase is absent or present only as a trace, which in a detonation problem is most of the domain for each phase. The upstream treatment and the first version of this work reset dilute phases ($\alpha < 0.01$) towards reference states and wrote the result back to the conserved variables. This destroyed energy (1–2% in the 0D transfer test) and removed the energy of the products expanding into the air. The following changes were made [THIS WORK]; together they make the solver conservative and keep trace phases bounded.

- *Conservative decode.* The conserved variables $\alpha\rho$, $\alpha\rho\mathbf{u}$, $\alpha\rho E$ are modified only where a phase is residual ($\alpha \le$ `residualAlpha`, 10^{-6}); bounds on velocity and temperature act on the primitive variables only. The decoded density is no longer bounded below by a reference density, which had held expanded gas at ambient density and overstated its pressure up to twenty-fold.
- *Pressure of trace phases.* Without pressure relaxation, all fluid phases in a cell undergo the same volumetric strain, so a trace of one fluid in another acquires an unrelated density and pressure. Non-carrier fluid phases are brought to pressure equilibrium with the carrier by a damped Newton volume exchange, $\Delta\alpha = (p_k - p_c)/(\rho_k c_k^2/\alpha_k + \rho_c c_c^2/\alpha_c)$, with the work $\bar{p}\,\Delta\alpha$ moved between the phase energies. Trial states outside the range of an EOS (e.g. gas compressed past its covolume) are rejected. In numerically mixed products/air cells the temperatures are relaxed with a stiff heat transfer integrated exactly; without it, products that are a trace by mass reached 10^5 K.

- *Energy of the relaxation and drag.* The instantaneous velocity relaxation added the dissipated kinetic energy to the total energy a second time; it now keeps each phase's internal energy plus its own dissipation. The work of the implicit drag is evaluated at the interfacial velocity averaged over the update.

- *Trace phases in the Riemann fluxes.* A phase that is a trace on one side of a face was given its own impedance in the HLLC flux. For a trace of explosive in an air cell this impedance is that of the explosive, and a small velocity of the air caused large face-pressure changes that destabilised the charge/air interface. Where a fluid phase is a trace on a side ($\alpha < \texttt{alphaTrace} = 10^{-3}$, blended linearly), the wave speeds and the common star pressure now use the mixture density, sound speed, velocity and pressure of that side, while the fluxes still carry the phase's own mass and energy; a residual upwind side carries no flux. A stationary unreacted charge next to air then stays at rest (Section 9.5).

- *Energy of the particle coupling and of the implicit stage.* A closed-domain energy check found three errors, all in pre-existing code. (i) The drag ODE solver moved momentum between the phases without the corresponding energy, so the frictional dissipation was lost (Houim 2: -0.55%). Each total-energy phase now receives $\Delta M \cdot U_I$ with U_I the update-averaged particle velocity, as in the explicit drag update. (ii) The particles take the pressure force $-\alpha_s \nabla p$, but the fluid energy equations did not supply the matching work: in an expanding flow the pressure work on the particle-occupied volume, $\alpha_s p\, \nabla \cdot \mathbf{u}$, was created (up to $+2.8\%$ in the spherical charge). The fluids now receive together $-\nabla \cdot (\alpha_s p\, \mathbf{u}_s) + \alpha_s \mathbf{u}_s \cdot \nabla p$, shared by volume fraction, which closes the total-energy balance with the particle momentum equation. (iii) The implicit stage (viscous and conductive terms) rebuilt the conserved fields from the bounded primitive fields and re-applied the difference between the previous step's conserved and primitive fields, which removed energy wherever the decode bounds were active (up to -1.2%). The implicit equations are now corrections to the current state with incremental updates of the conserved fields.

- *Residual stabilisation.* The upstream fluid momentum and energy equations added their residual-cell stabilisation with the wrong sign, which lowered the matrix diagonal to $(\alpha\rho - \alpha_{\mathrm{res}}\rho_{\mathrm{res}})/\Delta t$: zero where α equals the residual fraction. This caused the floating-point exceptions of the 2D Comp-B/Al tutorial at 116 and 171 μs; with the sign corrected the run reaches 200 μs.

- *Granular phase.* The particle packing limit is enforced by the frictional stress (with a small `alphaDeltaMin`), not by removing mass; the buoyancy on the particles uses the mixture fluid pressure, since the individual fluid pressures are undefined where a phase is absent.

The remaining known defects at this level are a few time steps in which trace gas ($\alpha \sim 10^{-6}$) inside the products reaches the 50 000 K decode cap, which is energetically negligible but not understood, (Section 10).

8 Verification Approach

8.1 Verification and validation

Verification checks that the equations are solved correctly: conservation, agreement with exact or reference solutions, and agreement with another code solving the same model. *Validation* checks that the model describes reality, by comparison with experiments. Most of the work so far is verification. The only comparison with empirical data is with the Kinney–Graham correlations for TNT, for a C4 charge (Section 9.3); no experimental data for aluminised charges have been used yet.

8.2 Test hierarchy and automation

Each model is tested in the simplest configuration that isolates it, from 0D reactors (one cell, no transport) through 1D planar problems to 1D spherical blasts; multi-dimensional cases follow. Every test has a verification script with explicit pass criteria. The runner `bin/run_blast_tests.py` [THIS WORK] runs the 18 tests in four tiers (Table 3), checks the solver logs for errors and for normal completion, and runs the scripts. Some tests depend on external builds (`EuDetCode`, `gunIB1D`).

Three limitations of the current verification should be noted. First, the programmed burn imposes the detonation velocity, so agreement of the front speed with D_{CJ} only checks the setup. Second, no

mesh-convergence studies have been made; the reported values are for one mesh per case. Third, several tests check reference models outside the solver (the aluminium and high-temperature air models) and are marked as such.

Table 3: Automated tests: what each checks, and the result of the most recent run. "Model" marks a standalone reference model that does not run the solver.

Tier	Test	What is checked	Result	Status
0	`EuDetCode` unit tests	CJ, ZND, isentrope calculations of the reference code	169/169	Pass
0	EOS fitter	Pressure continuity of the fitted EOS at 100 bar	$\Delta p = 0$	Pass
1	`phaseTransfer_0D`, `_3phase`	0D products → gas transfer (`blastEulerFoam`), with and without inert Al: conservation, pressure, temperature	Energy 4.2×10^{-9} / 2.2×10^{-9}, $\Delta p/p = 3 \times 10^{-7}$, 99.4% converted	Pass
1	Al particle kinetics	Heating, melting, ignition, Beckstead burning, oxide condensation	Reference behaviour reproduced	Model
1	`alMelting_0D`	Al heated through melting (`eMelting`, `blastEulerFoam`) vs an independent ODE solution	Plateau start and end within 0.1%, energy 4×10^{-9}	Pass
1	`particleDrag_0D`	Slip relaxation with the Loth drag: 100 μm at Ma 1.7, 1 μm at Kn 0.33, vs an independent solution	Within 0.03% of U_0	Pass
1	`particleHeating_0D`	Heating with the Kavanau Nusselt number, without and with slip, vs an independent solution	T_{Al} within 0.05%	Pass
1	`alBurning_0D`	Beckstead burning of 1, 10, 50 μm Al in air; no burning below T_{ign}	Burn time within 0.23% of the correlation, mass vs $(d/d_0)^3$ within 0.04%	Pass
1	High-T air	NASA-9 thermodynamics and O_2/N_2 dissociation equilibrium to 6000 K	Agreement with tabulated data	Model
1	NASA-9 thermodynamics	Compiled `nasa9` thermo, 18 species and air, 200–25 000 K, against the source data	$\leq 4.6 \times 10^{-6}$ (fitted condensed species and AlO: $\leq 2.7\%$ vs source)	Pass
1	Chemical equilibrium	Equilibrium at fixed (e, ρ): air, products, products/air, 0.01–100 kg/m^3, 2500–20 000 K, against `EuDetCode` and an independent solution	T, p within 4×10^{-4} of `EuDetCode` (its own tolerance), 10^{-6} of the independent solution	Pass
1	Equilibrium gas chemistry	Equilibrium above T_{eq} in the chemistry model: single cell (`blastFoam`) and 0D transfer (`blastEulerFoam`)	Composition, T, mass, elements, energy to round-off; energy incl. formation enthalpy 4×10^{-7}	Pass
1	Gas mechanism	Reversible GRI-3.0 $H_2/O_2/CO/C1$ subset in a closed cell (`blastFoam`) vs full GRI-3.0 (Cantera): product/air mixtures, 1200–2000 K, 1–50 bar	Ignition times within 0.94–1.13; final T within 0.8%	Pass
1	`pressurePhaseChangeTest`	1D transfer with species allocation (`blastFoam`)	Runs to completion	Pass
2	`ChargeCompBAl_1D` (tutorial)	Planar Comp-B/Al detonation and breakout into air	Runs to completion; air at 24 800 K (no dissociation)	Pass
2	`shock_dissociating_air`	Mach 13.9 shock in NASA-9 air (N_2 O_2 NO N O) in local equilibrium: exact equilibrium jump for the same data	p_2, T_2, ρ_2, u_2 within 1.7% (frozen: T_2 36% higher)	Pass
2	`multiphase_advection`	Contact advection, four flux schemes	Position < 1 cell; $\Delta p/p < 0.8\%$ (target 10^{-6})	Pass
2	`gunib1d_houim1-3`	Granular shock problems against `gunIB1D`	Within 1.2% (shock speed), 0.6% (peak pressure)	Pass
2	`planar_znd`	Planar Comp-B detonation: front position, CJ state, Taylor wave	Front at the programmed-burn position; CJ point of the products EOS at the burn speed (0.000%); front pressure 22.8–24.9 GPa (p_{CJ} 24.7 GPa)	Pass
2	`planar_znd_phaseTransfer`	Transfer behind a planar detonation	Smooth conversion below 100 bar	Pass
3	`spherical_blast`	1 kg C4 (`blastFoam`) against Kinney–Graham (TNT)	Peak overpressure 9–20% low; impulse 1–32%	Pass[a]

Tier	Test	What is checked	Result	Status
3	`tnt_blast`, `compB_blast`	1 kg TNT and Comp-B (`blastFoam`, 4 mm) against Kingery–Bulmash; mesh study to 1 mm	4 mm: overpressure 5–15% (TNT) and 2–10% (Comp-B) low for $Z \geq 1$; 1 mm TNT within 3.4% for $Z \geq 4$; impulse 14–17% (TNT), 5–9% (Comp-B) low	Pass[a]
3	`ChargeCompBAl_1D_spherical`	Spherical Comp-B/Al, inert Al	Peak 16.7 GPa, breakout 2.95 GPa; total energy conserved to 4×10^{-4}	Pass
3	`..._reacting`	As above with products → gas transfer and gas chemistry (NASA-9 gas, equilibrium above 2500 K, GRI-3.0 subset below)	12% of the products transferred by 35 μs; shocked air $\leq$ 9300 K; energy incl. formation enthalpy to 8×10^{-4}	Pass
3	`..._alCombustion`	As above with Al burning	No burning in the products; 2.5×10^{-6} of the Al burnt; energy to 8×10^{-4}	Pass
3	`..._interface`	Unreacted charge next to air stays at rest	< 0.001 m/s, mass and energy 1.5×10^{-9}	Pass

[a] The pass criteria are monotonic decay, overpressure within 25% and impulse within 40% per station; these do not meet the 5% target of the plan. [b] Expected until the products → gas transfer is enabled (Section 10).

9 Results

9.1 0D: transfer of products to gas

A single closed cell is initialised at the matching state (100 bar, $\rho = 35.234\,\mathrm{kg/m^3}$, 836.10 K) with eJWLC products, which transfer to the multicomponent gas with $\tau = 20\,\mu$s; the three-phase variant adds inert aluminium (Figure 4). After 99.4% of the products mass has transferred, the total energy is conserved to 4.2×10^{-9} (2.2×10^{-9} with aluminium), the mass to 2.4×10^{-9}, and the net momentum is zero. The gas forms at the products temperature (difference $< 10^{-3}$ K) and the pressure stays constant to 3×10^{-7}. This confirms the conservation of the transfer and the datum reconciliation of Section 4.4. With the earlier datum convention ($\int c_v\,\mathrm{d}T$ from 298.15 K) the pressure rose by 0.58%; with the conservative decode not yet in place, the reset of the dilute products removed 1.8% of the energy. The isobaric variant of the test planned in Stage 1 has not been made.

9.2 1D canonical problems

Shock in air. A driver at 1700 bar and 7500 K drives a shock into air at 1 atm and 300 K (400 cells, 1 m, `blastFoam`, HLLC). The air is N_2, O_2, NO, N and O with NASA-9 thermodynamics (perfect gas) and local chemical equilibrium above 2500 K, with no finite-rate reactions. The shock moves at 4839 m/s (Mach 13.9), measured by a fit of the sub-cell shock position over the last three writes, and the solver gives $p_2 = 244$ bar, $T_2 = 6626$ K, $\rho_2 = 10.95\,\mathrm{kg/m^3}$ and $u_2 = 4310$ m/s (Figure 5). At the measured shock speed the exact jump for the same species data in chemical equilibrium (computed independently of the solver) gives $p_2 = 247$ bar, $T_2 = 6603$ K, $\rho_2 = 11.15\,\mathrm{kg/m^3}$ and $u_2 = 4330$ m/s: the solver is within 1.2%, 0.35%, 1.7% and 0.46%, and its post-shock composition ($Y_\mathrm{O} = 0.162$, $Y_\mathrm{NO} = 0.096$, $Y_\mathrm{N} = 0.017$, $Y_{\mathrm{O}_2} = 0.019$) is within 0.0011 of the equilibrium values. Without the chemistry (frozen air) the same shock would give $T_2 = 8990$ K and $\rho_2 = 9.30\,\mathrm{kg/m^3}$; before this work, with the JANAF air extrapolated without dissociation, the solver gave about 10 000 K at Mach 14.5. The driver gas is also in equilibrium, which weakens the shock from Mach 14.5 to 13.9.

Contact advection. A density contact (116 / 16 $\mathrm{kg/m^3}$) advected at 500 m/s and 100 bar stays within one cell of its exact position for HLLC, AUSM+, Kurganov and Tadmor (Figure 6). Spurious pressure oscillations at the contact are 4.0–4.4% with the van Leer limiter and below 0.8% with Minmod, which is why Minmod is used; the target of 10^{-6} is not met, as expected of a scheme that is not designed to be pressure-equilibrium-preserving.

Granular shock problems. Three problems of Houim and Oran type (advection of a particle curtain, a shock passing through a particle bed, and compaction of a dense bed) are solved with `blastEulerFoam` and with `gunIB1D` (Figure 7). The curtain advection is identical; for the shock tube the transmitted

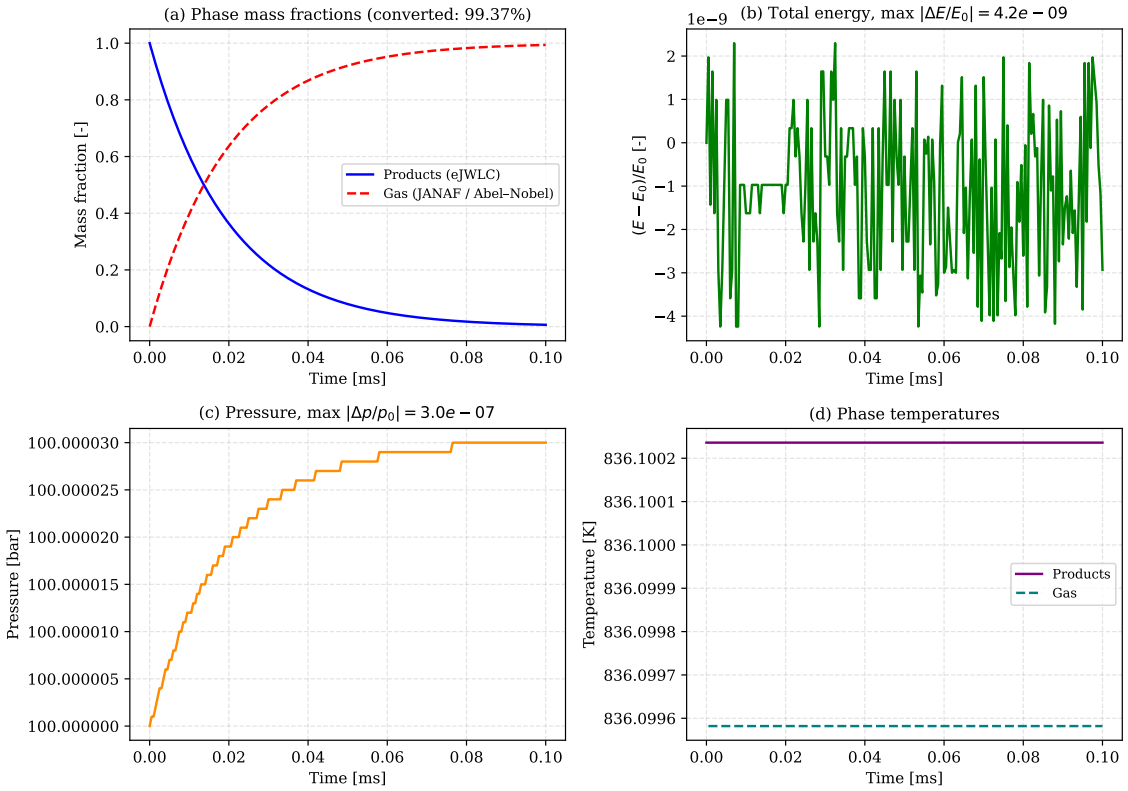


Figure 4: 0D products → gas transfer in `blastEulerFoam` at 100 bar: (a) phase mass fractions; (b) relative total energy error ($\leq 4.2 \times 10^{-9}$); (c) pressure; (d) phase temperatures.

shock speed differs by 1.20% (489.2 vs 495.1 m/s), the post-shock pressure by 0.62%, the gas velocity by 0.25% and the particle velocity by 0.32% (2.1% before the drag energy fix of Section 7.2); the total energy is conserved to 6×10^{-7}; for the compaction problem the peak pressure differs by 0.59%, and the compaction wave speed of `blastEulerFoam` (183.8 m/s) is within 0.8% of the reference value. The compaction result required the frictional stress to act (`alphaDeltaMin` 10^{-6}); with the upstream tutorial value the bed packed to the maximum fraction and the old decode removed 41% of the particle mass.

Planar detonation. A planar Comp-B charge (without aluminium, 1717 kg/m^3) is initiated at a closed end (`blastFoam`, 400 cells, 0.5 mm, SSP Runge–Kutta (RK2SSP) time integration at a Courant number of 0.35). The products EOS is the `EuDetCode` JWL fit for Comp-B (BKW), with E_0 chosen so that the reference isentrope passes through the CJ point for the reactant energy of the case; the CJ point of this products EOS (the minimum speed on its Hugoniot) is then the programmed-burn speed, 7934.3 m/s. The pressure at the burn front stays at 22.8–24.9 GPa ($p_{\mathrm{CJ}} = 24.66$ GPa) and the solution behind follows the Taylor expansion (Figure 8). The earlier setup used the LLNL Comp-B JWL with $E_0 = -8.5$ MJ/kg (the LLNL value is 8.5 GJ/m^3, i.e. 4.94 MJ/kg) and the `EuDetCode` burn speed of 7947 m/s: the CJ speed of that products EOS is 8403 m/s, so no steady detonation exists at the imposed speed, the energy released behind the front overdrives it, and the pressure at the front grew to 55, 95 and 135 GPa at 5, 10 and 15 μs. The test now checks the consistency of the products EOS with the burn speed and the front pressure. The variant with `pressurePhaseChange` converts the products to gas below 100 bar without disturbing the wave.

9.3 Spherical blast of a C4 charge

A 1 kg spherical C4 charge (JWL, programmed burn at the CJ speed of the JWL, 8227.6 m/s; `blastFoam`) is detonated at the centre of a 1D spherical domain of 6 m with 300 cells ($\Delta r = 20$ mm), and the peak overpressure and positive impulse are recorded in every cell with `temporalMax` and `temporalIntegral` (Figure 9). The reference is the Kinney–Graham correlation for TNT [10] (the verification script is named after Kingery–Bulmash [8], but evaluates the Kinney–Graham formulas; the C4 mass is used

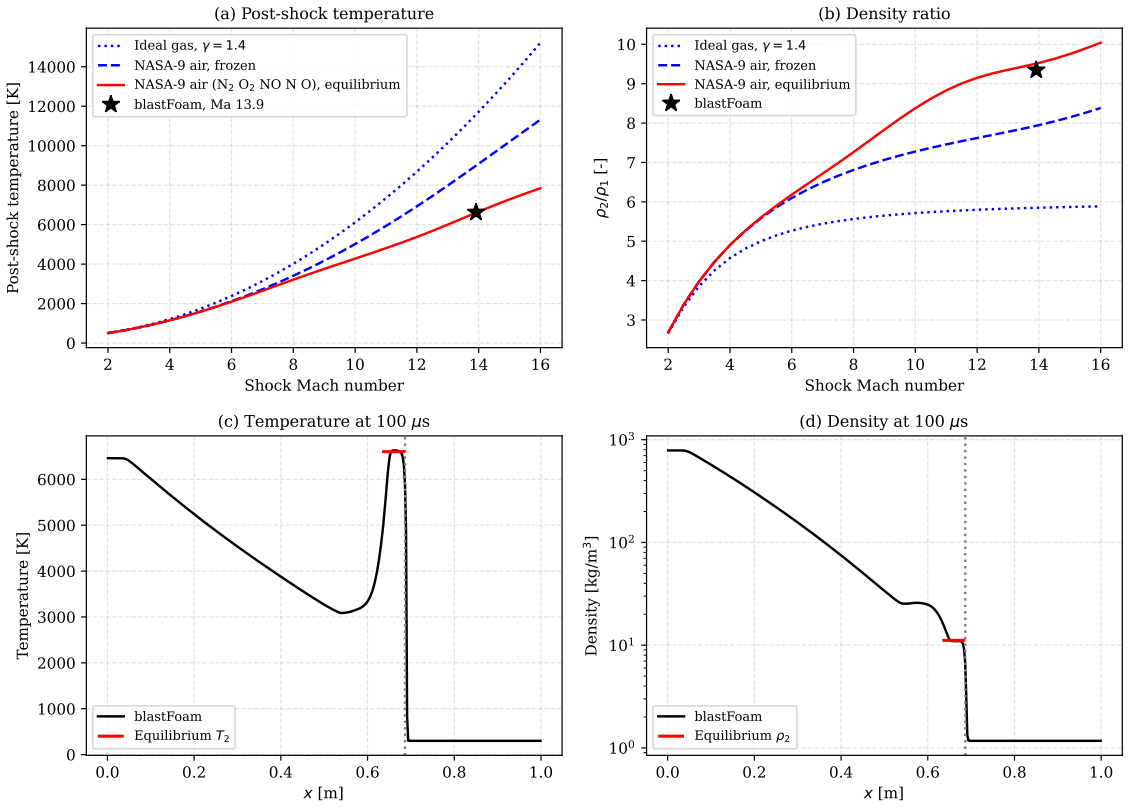


Figure 5: Mach 13.9 shock in air: (a) post-shock temperature and (b) density ratio against shock Mach number for an ideal gas and for NASA-9 air (N_2 O_2 NO N O), frozen and in chemical equilibrium (exact jumps), with the `blastFoam` result (equilibrium chemistry above 2500 K); (c) temperature and (d) density at 100 μs, with the equilibrium post-shock values over the shocked layer.

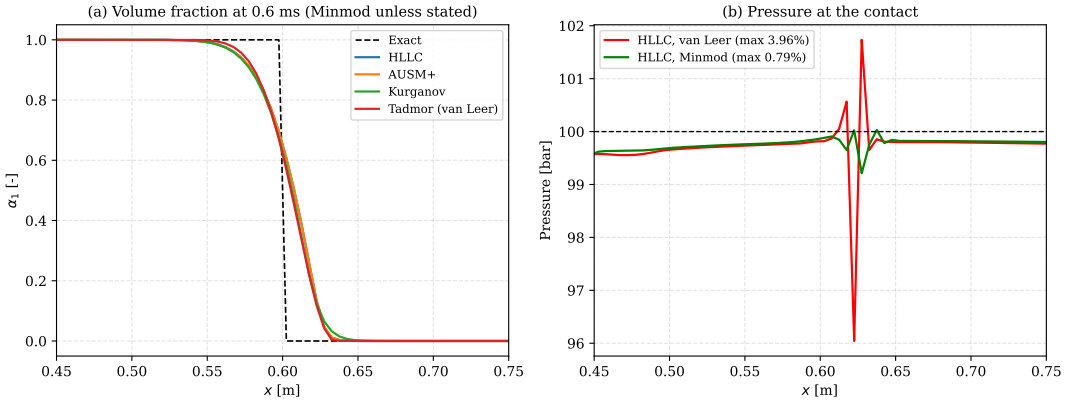


Figure 6: Contact advection at 500 m/s, 100 bar: (a) volume fraction at 0.6 ms for four flux schemes; (b) pressure at the contact with HLLC and the van Leer and Minmod limiters.

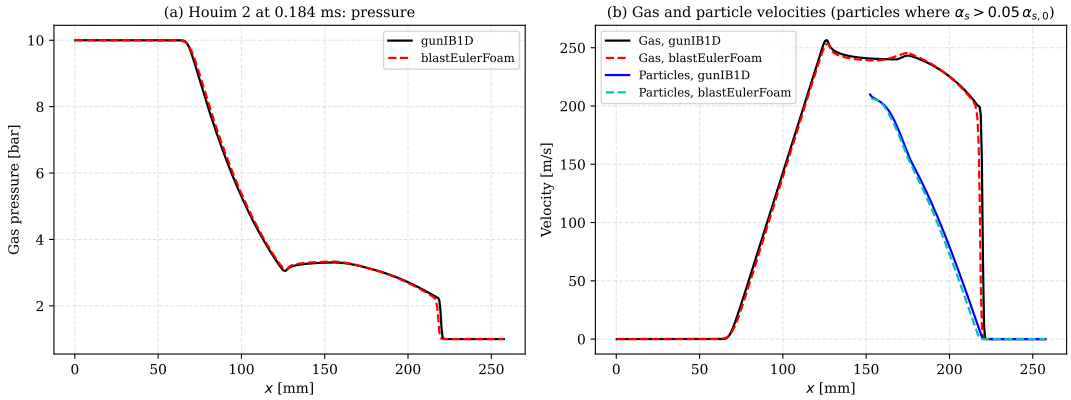


Figure 7: Granular shock tube (Houim 2) at 0.184 ms, `blastEulerFoam` against `gunIB1D`: (a) gas pressure; (b) gas and particle velocities, the particle velocity where the particle volume fraction exceeds 5% of the initial bed value (`gunIB1D` keeps a residual fraction of about 2% with the gas velocity where the particles have left, `blastEulerFoam` sets the velocity of a residual phase to zero).

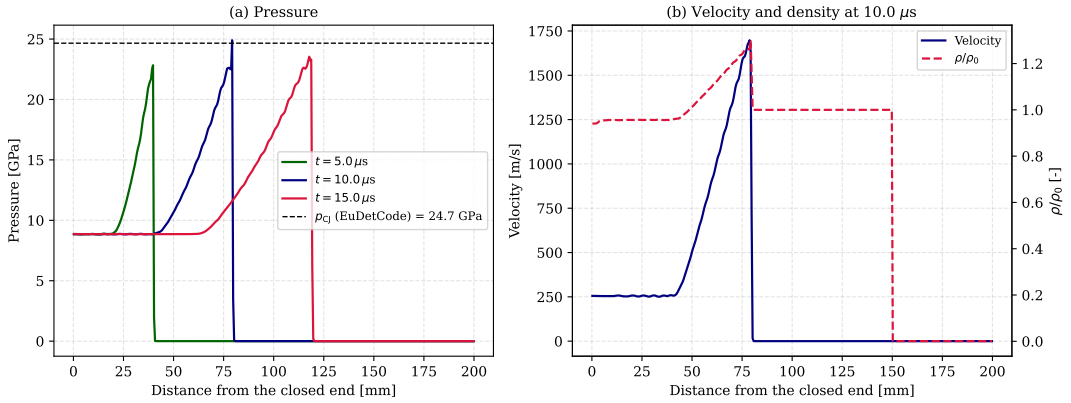


Figure 8: Planar Comp-B detonation (`blastFoam`): (a) pressure at 5, 10 and 15 μs; (b) velocity and density at 10 μs.

without TNT equivalence). The peak overpressure is low at all scaled distances: 36.2 vs 40.0 bar at $Z = 0.5\,\mathrm{m/kg^{1/3}}$, 8.5 vs 10.1 bar at $Z = 1$, 1.71 vs 2.08 bar at $Z = 2$ and 0.75 vs 0.82 bar at $Z = 3$ (9–20%). The positive impulse differs by 28%, 32%, 2%, 1% and 9% at $Z = 0.5$, 1, 1.5, 2 and 3 (until 2026-10-15 the script's Kinney–Graham impulse integrated a Friedlander pulse with an assumed decay parameter, which underestimated the far-field impulse by a factor of up to 3; it now uses the published closed form). (With the earlier burn speed of 7850 m/s, below the CJ speed of the JWL, the peaks were about 1% lower.) Since C4 is more energetic than TNT (TNT equivalence about 1.2–1.4), the simulated overpressures would be expected to lie above the TNT curve; the low peaks therefore point to the resolution (20 mm cells smear the shock) and to the JWL parameters of the case. A mesh-convergence study and a TNT or Comp-B case are needed before the 5% target of the plan can be assessed.

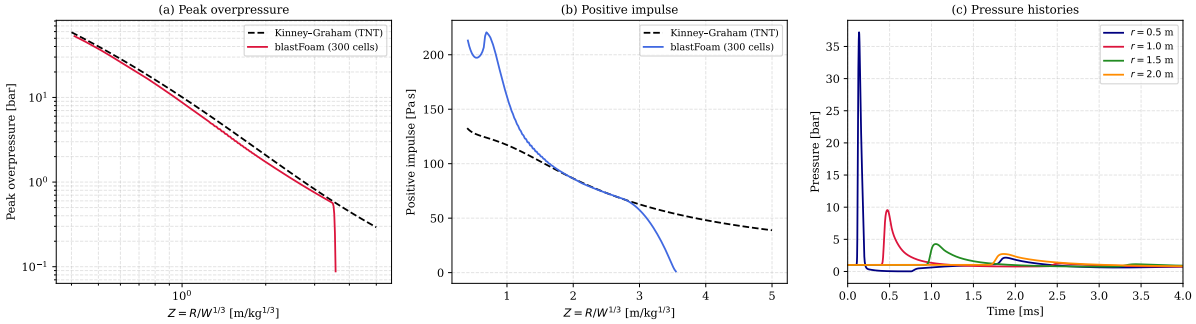


Figure 9: 1 kg C4 spherical blast: (a) peak overpressure against scaled distance; (b) positive impulse, with the Kinney–Graham correlations for TNT; (c) pressure histories at 0.5–2 m.

9.4 Spherical TNT and Comp-B charges against Kingery–Bulmash

Plan step E1 compares 1 kg spherical charges of TNT and of aluminium-free Comp-B with the Kingery–Bulmash data for TNT [8] (Figure 10). The cases (`tests/1D_spherical/tnt_blast`, `compB_blast`; `blastFoam`, 1D spherical sector of 10 m, centre-initiated programmed burn at the CJ speed of the products EOS, ideal-gas air at 101 325 Pa and 288.15 K, no afterburning) use `EuDetCode` BKW products: TNT at 1630 kg/m^3 ($D = 6696$ m/s, $p_{CJ} = 16.9$ GPa; `utilities/thermo_fitter/eudet_tnt`) and the Comp-B of the planar test. The reference (`tests/1D_spherical/kingery_bulmash.py`) is the Kingery–Bulmash free-air overpressure polynomial; for the impulse and the arrival time, for which only the hemispherical fits of Swisdak [9] were available, the hemispherical value at the charge mass that reproduces the free-air overpressure at the same scaled distance is used (a mass ratio of 1.3–2.1, instead of the usual constant reflection factor 1.8, which would shift the overpressure by −13 to +7%). The Swisdak coefficients reproduce the upstream validation data and the examples of IATG 01.80 within 1%. Comp-B is compared through the TNT equivalence of UFC 3-340-02 (1.11 for pressure, 0.98 for impulse).

The arrival times agree within 1.3% for $Z \geq 2\,\mathrm{m/kg^{1/3}}$. The peak overpressure depends on the mesh, and on the resolution of the charge more than of the air: for TNT at $Z = 8$ it is 5.6, 4.1 and 0.5% below the reference with 4, 2 and 1 mm cells, and 1.4% below with 1 mm cells only to 0.3 m (2 mm beyond); the limiter (Minmod or van Leer) changes it by less than 0.5%. At 1 mm the TNT peak overpressure is within 3.4% of the reference for $Z \geq 4$ and 5–8% low at $Z = 1.5$–3; Comp-B at 2 mm is within 4% for $Z \geq 4$ and within 8% for $Z = 1$–3. Gate 3 (5% for $Z \geq 1$) is therefore met in the far field but not at $Z = 1.5$–3, where also the reference is least certain. The positive impulse does not depend on the mesh and is low: for TNT 14–17% at $Z = 4$–8 (Kinney–Graham [10] gives similar far-field values), for Comp-B 5–9%. Since the overpressure agrees there, the deficit is a short positive phase (TNT: 13–23% shorter than the reference at 4–8 m); the larger deficit for TNT, whose products are much more oxygen-deficient (oxygen balance −74% against −43% for Comp-B), points to the afterburning of the products in air, which is part of the experiments but not of these cases. The tests run on the 4 mm mesh (8–13 min) with regression bounds (arrival 3% for $Z \geq 3$, overpressure 20% and impulse 25%).

9.5 Spherical Comp-B/Al charge

A spherical Comp-B/Al charge of radius 50 mm (0.95 kg) is initiated at its centre in a 1D spherical domain of 0.2 m with 400 cells ($\Delta r = 0.5$ mm) and run to 40 μs with `blastEulerFoam`. Three variants share the

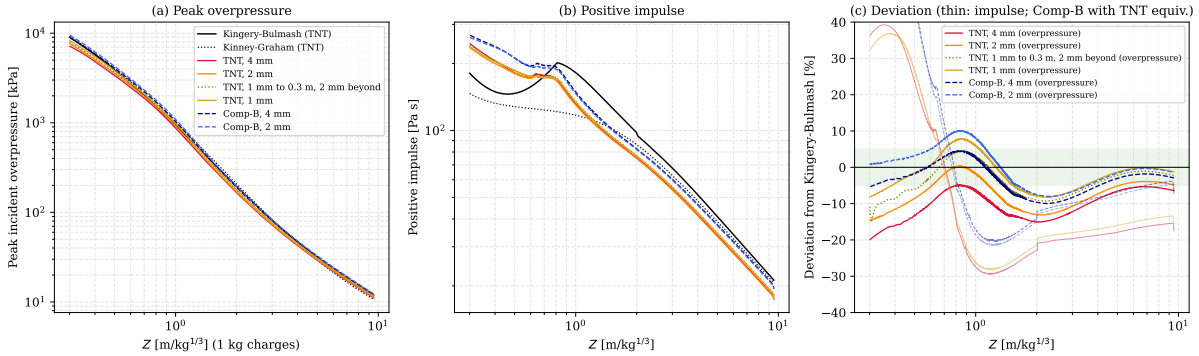


Figure 10: 1 kg TNT and Comp-B free-air blast against Kingery–Bulmash (TNT; Comp-B with TNT equivalence): (a) peak overpressure, (b) positive impulse (radial `temporalIntegral`, which includes the slight overpressure remaining near the charge; the verification uses the first positive phase), (c) deviation from the reference (thick: overpressure, thin: impulse), for the meshes of the study. The kinks of the reference impulse at $Z \approx 0.8$ and 2 are the range limits of the hemispherical fits; at $Z \leq 1$ the impulse references differ by up to 50% (Kingery–Bulmash and Kinney–Graham).

thermodynamics: inert aluminium, the same with gas-phase chemistry, and the same with aluminium burning (Figure 11).

- *Detonation.* The front moves at 7163 m/s, as imposed. The peak pressure grows to 16.7 GPa at the charge edge, 83% of p_{CJ}; the CJ spike is not resolved on this mesh, and the effect of resolution has not been quantified.

- *Breakout and air shock.* In the reacting cases (NASA-9 gas in local equilibrium above 2500 K) the pressure drops to 3.09 GPa at the charge/air interface, and the air shock is 43 MPa at 80 mm, 33 MPa at 100 mm (arrival 14.2 μs) and 20.7 MPa at 150 mm; with the earlier JANAF gas without dissociation it was 45, 34 and 21 MPa. These values changed substantially during the work (e.g. 0.16 GPa at 100 mm before the density floor was removed, 29 MPa before the constant-c_p extrapolation, 35 MPa before the energy fixes below), which shows the sensitivity of the near field to the treatment of trace phases, high-temperature air and energy conservation. With dissociation the shocked air reaches at most 9300 K (at 10 μs), against 16 500 K without it. The first mixed cell at the charge/air interface still reaches the 50 000 K decode cap during breakout (6.9–7.2 μs): its gas stays at ambient density while the products are at 0.7 GPa, a pressure non-equilibrium of the mixed cell (Section 10), not of the chemistry.

- *Aluminium.* Drag accelerates the aluminium to 1910–2250 m/s where $\alpha_{\mathrm{Al}} > 10^{-3}$ (the higher value in the inert case, which applies drag also to the dilute edge of the cloud). The particles are heated to at most 540 K by 40 μs, far below ignition. With the burn rate limited to gas-phase oxidiser, there is almost no burning inside the products (diameter change $\leq 0.0043\%$); in 1D the particle cloud stays inside the products except at its dilute edge, and 9.4×10^{-5} of the aluminium mass burns in 40 μs.

- *Products afterburning.* In the reacting and Al-combustion variants the products are transferred to the gas below 100 bar ($\tau = 20\,\mu$s). By 35 μs 12.4% of the products mass has been transferred; the carbon in the gas is then still distributed as in the transferred products (53% CO, 18% CO_2, 9% CH_4, 20% C(gr)), since in 1D the transferred products and the air hardly mix in this time. The blast pressures to 40 μs are unchanged by the transfer, which happens behind the shock; the afterburning becomes significant only on longer times and larger domains.

- *Interface.* A separate case with an unreacted charge next to air stays at rest: velocities below 10^{-3} m/s at 5 μs and below 1.4 mm/s for 40 μs, pressure at 1.013 bar, mass and energy conserved to 1.5×10^{-9}. Before the trace-phase changes of Section 7.2 this case became unstable within microseconds.

These cases show that the detonation, breakout and early air shock of a three-phase charge run stably. While the domain is closed (after the burn, before the shock leaves), the total energy of the three phases, including the kinetic and granular energy of the Al, is conserved to 4×10^{-4} (inert) and 8×10^{-4} (with

transfer and chemistry, counting the formation enthalpy of the gas), and the Houim 2 gas/particle shock tube to 6×10^{-7}; the tests check this. Before the fixes of Section 7.2 the spherical cases gained up to 2.8% and then lost 1.2%, and Houim 2 lost 0.55%. The products transfer adds a mass error of up to 5×10^{-6} (transfer into cells where the gas is still residual). They do not yet test the afterburning, which needs the products transfer, the aluminium ignition and combustion models, a domain of metres and run times of milliseconds.

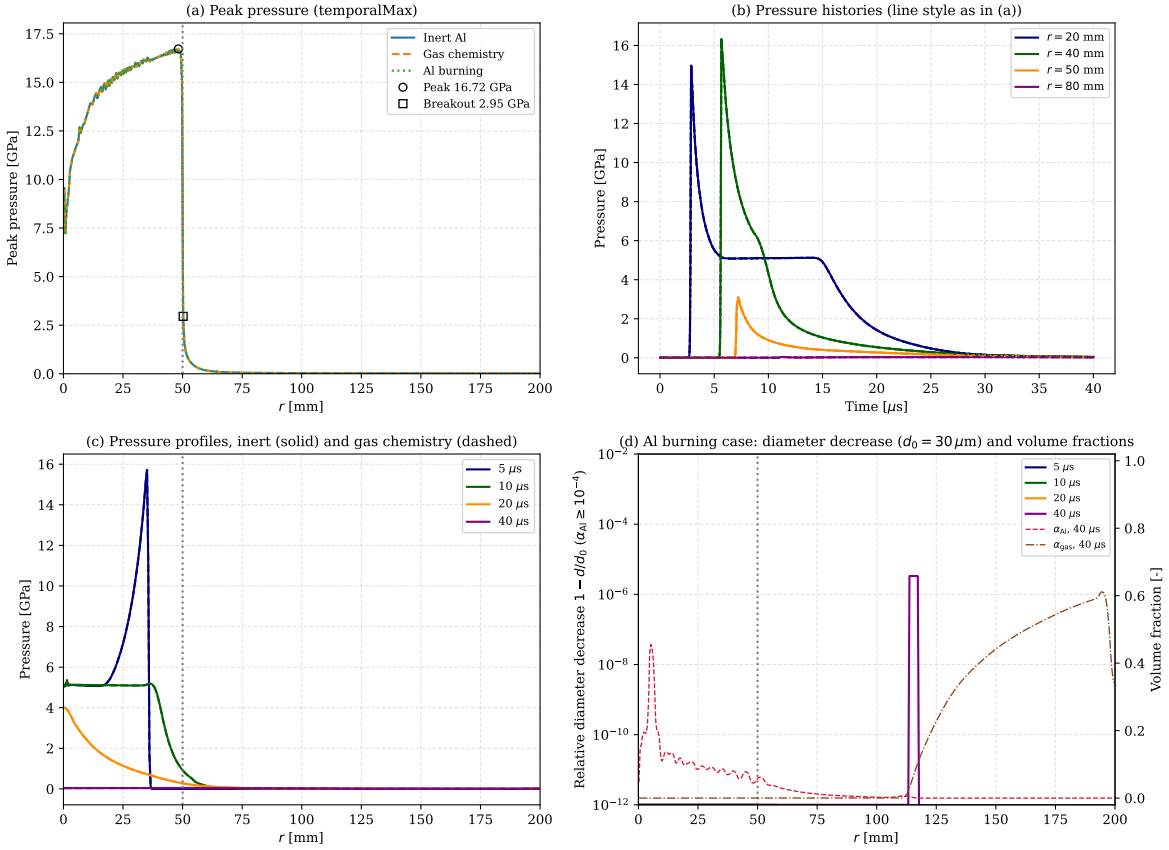


Figure 11: Spherical Comp-B/Al charge in `blastEulerFoam` (inert Al; gas chemistry and Al burning, both with the products → gas transfer): (a) peak pressure against radius (`temporalMax`); (b) pressure histories at 20, 40, 50 and 80 mm; (c) pressure profiles at 5, 10, 20 and 40 μs; (d) particle diameter and volume fractions.

9.6 Aluminium particles

The standalone particle model (`tests/0D/verify_al_particle_kinetics.py`) integrates the heating of a single particle with an internal temperature profile (Goodman heat-balance integral, parabolic profile), melting, ignition at 2050 K, Beckstead burning and oxide condensation (Figure 13). For a 20 μm particle at $h = 3 \times 10^5$ W/(m^{2}K) in gas at 2800 K the surface reaches the ignition temperature at 32.9 μs, 15.5% earlier than with a uniform particle temperature (38.9 μs), and the Beckstead correlation gives a burn time of 1.6 ms in air. (Until 2026-10-14 the burn-time function of this model used a dimensionally inconsistent rate constant that gave burn times of seconds; it now evaluates the published correlation.) The model has not been compared with experimental ignition delays or burn times; it serves as the reference for the solver implementation.

Aluminium models in the solver. The models of Section 5.3 are verified in single-cell `blastEulerFoam` runs against independent integrations of the same model equations (Figure 12). (a) 30 μm particles at a volume fraction of 10^{-4}, heated by gas at 2500 K and 50 bar through a constant heat transfer coefficient, stay at the melting point from 0.098 to 0.169 ms, within 0.1% of the reference, and the total energy is conserved to 4×10^{-9}. (b) The slip velocity of 100 μm particles started at 600 m/s (Ma 1.7, compression

branch of the Loth drag) and of 1 μm particles at 0.2 bar (Re 5, Kn 0.33, rarefaction branch) follows the reference within 0.03% of the initial slip. (c) With the Kavanau Nusselt number a 1 μm particle at 0.2 bar heats with $Nu = 0.28$ instead of 2; the particle temperature follows the reference within 0.05%, also with 300 m/s slip and drag. (d) Ignited 1, 10 and 50 μm particles in air at 2500 K and 1 atm burn out in the Beckstead time (7.3 μs, 0.46 ms, 8.4 ms) within 0.23%, and the particle mass follows $(d/d_0)^3$ within 0.04%; particles at 1800 K, below $T_{\mathrm{ign}} = 2000$ K, do not burn.

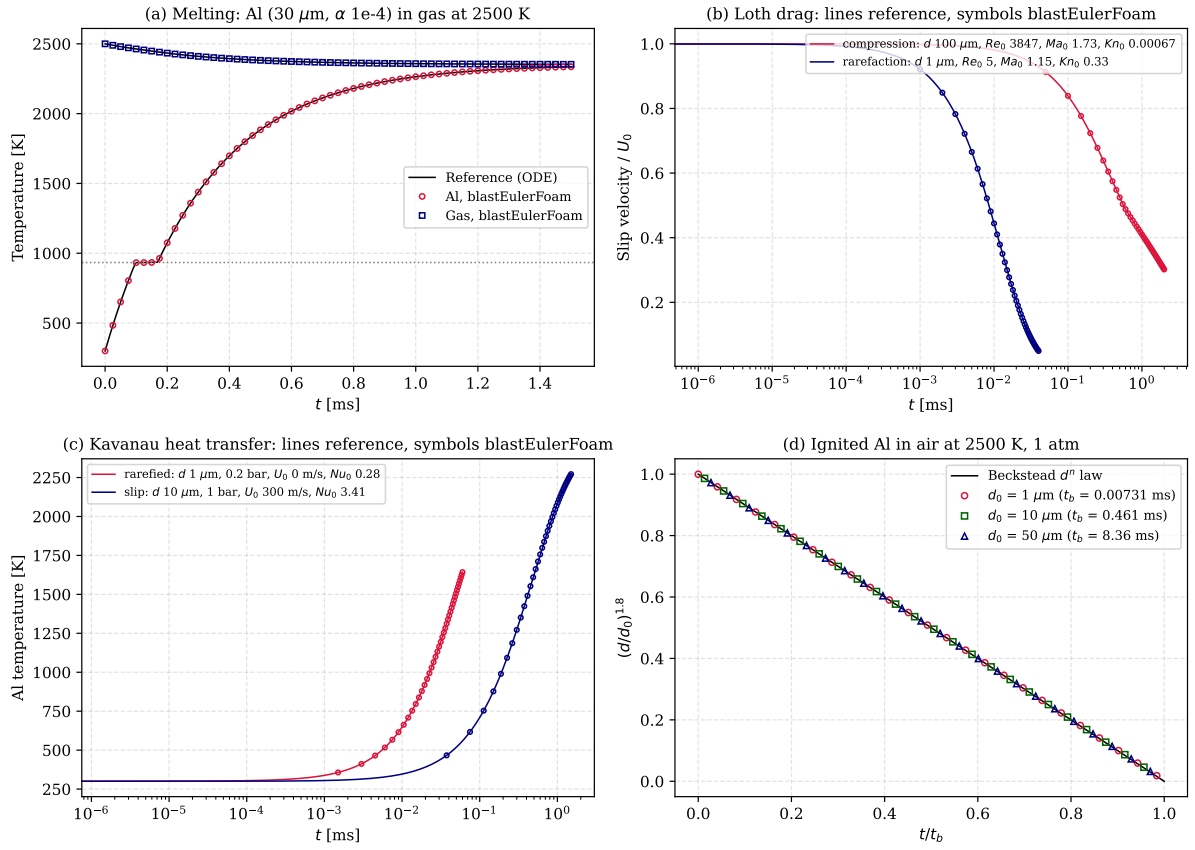


Figure 12: Aluminium particle models in `blastEulerFoam` (single cell) against independent solutions: (a) melting (`eMelting`); (b) slip relaxation with the Loth drag; (c) heating with the Kavanau Nusselt number; (d) Beckstead burning after ignition.

10 Status and Plan

10.1 Status against the aims

Table 4 summarises the status for the four parts of the problem.

10.2 Open issues

- Total energy in the spherical cases is conserved to 4–7 $\times\, 10^{-4}$, not yet to round-off (most of the error arises at the end of the programmed burn).
- Melting, the Loth drag, the Kavanau heat transfer, the ignition criterion and the Beckstead rate are in the solver but used only in the E2 runs of the 2 m tutorial, not in the Comp-B/Al tests; oxide condensation is not in the solver. The reacting diameter model resets the diameter, and stops burning, below a volume fraction `alphaMin` (default 10^{-4}, reached at about 0.5 m in the spherical case).
- The first mixed cell at the charge/air interface reaches the 50 000 K decode cap during breakout (gas at ambient density next to products at 0.7 GPa); the inert cases and the tutorials still use the

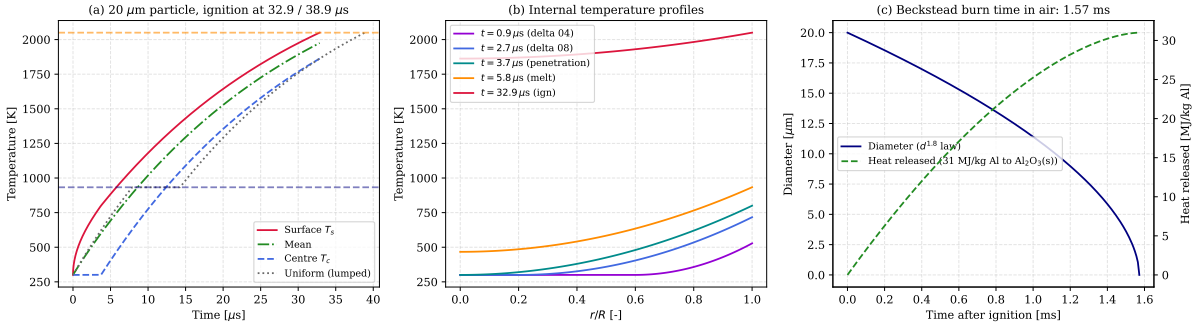


Figure 13: Reference particle model (not in the solver): (a) surface, mean, centre and uniform (lumped) temperatures of a 20 μm particle in gas at 2800 K; (b) internal temperature profiles; (c) diameter and heat release after ignition, with the Beckstead burn time in air.

Table 4: Status against the aims (October 2026).

Part	**In place and verified**	**Missing**
Detonation and early expansion	Products-only eJWLC fit; programmed burn; three-phase spherical charge, stable through breakout; stable charge/air interface	Resolution of the CJ state; dissociation in shocked air; mesh convergence
Products/air mixing and chemistry	Products → gas transfer (0D, planar); multicomponent reacting gas phase	Transfer enabled in the spherical cases (2026-10-14); conservation with particles; domains of metres and run times of milliseconds
Al heating, ignition, combustion	Granular Al phase; melting, Mach- and Knudsen-dependent drag and heat transfer, ignition and Beckstead burning in the solver (0D tests); empirical burn rate limited to gas-phase oxidiser in the cases	The new models in the Comp-B/Al cases; oxide condensation; experimental comparison
Afterburning in air	Gas-phase Al and CO/H_2 reactions in the cases	Multi-dimensional mixing; turbulence and turbulence–chemistry closures; radiation; validation

JANAF gas without dissociation.

- Blast against Kingery–Bulmash (Section 9.4): with 1 mm cells the TNT peak overpressure is within 3.4% for $Z \geq 4$ but 5–8% low at $Z = 1.5$–3, and the positive impulse is 14–17% low (Comp-B: 5–9%), most likely the missing afterburning of the products. The test meshes (4 mm) give 5–15% low overpressures.

- Transfer sensitivity (0.2 m case, first part of WP 3.2): for transfer pressures of 100–200 bar and $\tau \geq 1\,\mu$s the gauge peaks at 80/100/150 mm vary by 0.05/0.56/2.9%; on the 2 m case (100 bar/20 μs, 100 bar/2 μs, 200 bar/20 μs, step E2) the peaks at 0.3–1.9 m vary by at most 1.5% and the impulse at 0.3–1 m by 1.8%. A fast transfer at high pressure used to abort the run: in the burn front the reaction progress is above zero before the pressure has risen, and the transfer moved unreacted charge into the gas; the transfer is now limited to completed burn ($\lambda \geq 0.99$).

- Inert vs reacting Comp-B/Al (step E2; 2 m, 2 ms, with melting, Loth drag, Kavanau heat transfer and Beckstead burning of the Al, and the Al sub-oxides and liquid Al_2O_3 in the gas): the products → gas transfer raises the positive impulse by 3.4% at 0.5 m and 4.5% at 1 m and the peak overpressure by at most 2.9%; in 1D the products mix little with the air. The aluminium does not ignite: by 2 ms it has only reached its melting point (17% of the mass on the melting plateau, none near the 2050 K ignition temperature), so the run with Al burning equals the one without, and the aluminium contribution to the impulse remains to be shown.

- The 2D axisymmetric Comp-B/Al tutorial has reached 200 μs only from a restart (after the residual-stabilisation fix); a complete run from the start remains to be made. A few steps of trace gas reach the temperature cap.

- Programmed burn needs a products EOS whose CJ speed is not above the burn speed (checked in the planar test). The Comp-B/Al cases run slightly faster than the CJ speed of their products EOS and sit on the weak branch (2.4% below p_{CJ}); the upstream C-4 tutorials ran at 7850 m/s against 8213 m/s for their JWL and were overdriven in the same way as the earlier planar case (front pressure growing to about 125 GPa in a planar test); their burn speed is now the CJ speed of the products EOS, and the reactants have a Murnaghan stiffness from the ingredient data (before: none). A checker (`utilities/thermo_fitter/check_programmed_burn.py`) compares the burn speed with the products EOS for any case. The planar tests used forward-Euler time integration with second-order reconstruction, which is not TVD-stable; the resulting oscillations behind the front (rms 1–2% of p_{CJ}, growing with the Courant number) disappear with the SSP Runge–Kutta integrator used elsewhere (rms 0.2%).

- No mesh-convergence studies; no experimental data for an aluminised charge identified.

10.3 Plan

The plan (details in `report/status_review_2026-10-14.md`) is ordered by the path to the final validation:

A. *Consolidate*: documentation consistent with the code, verification scripts that fail on incomplete or non-reacting runs, one complete regression run, the 2D crash and the remaining trace-phase defects.

B. *Products afterburning*: enable the transfer in the spherical cases, extend them to 2 m and 2 ms, study the sensitivity to the transfer pressure and time (target: gauge pressures varying by less than 1.5% for 100–200 bar), and add a reduced $CO/H_2/O_2$ mechanism.

C. *High-temperature air*: NASA-9 thermodynamics to 20 000 K, an equilibrium solver at fixed (e, ρ) (verified against `EuDetCode`) and local equilibrium above T_{eq} in the gas chemistry are done, and the shock-in-air test matches the equilibrium jump (Section 9.2). A reversible finite-rate mechanism for use below T_{eq} (GRI-3.0 $H_2/O_2/CO/C1$ subset, 68 reactions, reverse rates from the NASA-9 data) agrees with the full GRI-3.0 mechanism in ignition (within 13%) and burnout (within 0.8% in temperature) for product/air mixtures at 1200–2000 K and 1–50 bar. The reacting Comp-B/Al cases use the NASA-9 gas, the equilibrium above T_{eq} and this mechanism; the shocked air stays below 9300 K.

D. *Aluminium in the solver*: melting, Mach- and Knudsen-dependent drag and heat transfer, and ignition with the Beckstead rate are done, each with a 0D solver test (Section 9.6); oxide condensation and the comparison with experimental ignition delays and burn times remain.

E. *1D validation*: TNT and aluminium-free Comp-B against the Kingery–Bulmash data with a mesh study are done (Section 9.4; Gate 3 met only for $Z \geq 4$, impulse low without afterburning); the inert and reacting Comp-B/Al charge give 3–5% more impulse from the products afterburning at 0.5–1 m and none from the aluminium, which does not ignite; next the aluminium ignition in the spherical case.

F. *Particle method*: compare the existing Eulerian and Lagrangian solvers on the same cloud problem; develop population-balance or hybrid methods only if this shows a need.

G. *Multi-dimensional*: 2D axisymmetric repeat of the spherical case with AMR, cylindrical charges, instability-driven mixing with LES and a turbulence–chemistry closure, and radiation where significant.

H. *Validation*: identify an experimental Comp-B/Al dataset now, then simulate the trial geometry (targets: arrival time within 3%, peak overpressure within 7%, positive impulse within 8%).

B, C and D are independent and can proceed in parallel after A; E needs B–D, and the multi-dimensional work needs E.

A Upstream Demonstration Cases

The following `blastFoam` tutorials [UPSTREAM] cover phenomena relevant to the problem but have not been assessed in this work: `shock_Air_Al_Air` (shock through a particle curtain), `reactingParticles` (`blastEulerFoam` shock tube with reacting particles; its time step collapses at 0.27 ms, not analysed), `multicomponent_shock_bubble` (Richtmyer–Meshkov instability), `twoChargeDetonation` (blast wave interaction and Mach reflection), `axisymmetricChargeMod` and `axisymmetricChargeReacting` (2D axisymmetric charge with AMR), and `multiphase_reacting_bubble` (detonation, transfer and afterburning in `blastFoam`). The `blastEulerFoam` tutorials `ChargeCompBAl`, `ChargeCompBAl_1D` and `StationaryCharge_1D` [THIS WORK] use the settings of the spherical test cases.

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